

JaeHyeok Han

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EDUCATION

The Catholic University of Korea

Mar. 2025 – Present

B.S. in Biomedical Software GPA: 4.22 / 4.50

RESEARCH INTERESTS

Clinical AI, Electronic Health Records, Robustness and Transportability,, Clinical Natural Language Processing, Large Language Model

RESEARCH EXPERIENCE

Undergraduate Researcher

November. 2025 – Present

The Catholic University of Korea Advisor: Prof. Hyo Jung Kim

- **Information-Channel Decomposition in Electronic Health Records**

Investigating how observation patterns, recorded values, and static patient information contribute to clinical prediction using MIMIC-IV and eICU.

- **Transportability of HPV Risk Prediction Models**

Studying model transportability from NHANES to multinational DHS populations under limited covariate overlap and label-free target settings.

- **Clinical Text Inference with Large Language Models**

Evaluating LLMs for clinical concept detection, encounter-status inference, and negation recognition.

- **Incisional Hernia Outcomes Using National Claims Data**

Contributing to cohort definition, exploratory analysis, subgroup design, and statistical analysis using Korean national claims data.

- **Breast Cancer Subtype Classification from Pathology Reports**

Conducted comparative NLP and LLM experiments for breast cancer subtype classification using IHC pathology reports.

- **FAERS Data Preprocessing for Pharmacovigilance Research**

Contributed to preprocessing and deduplication of FDA Adverse Event Reporting System data for pharmacovigilance analysis.

CONFERENCE SUBMISSIONS

- **Han, J.**, et al. “Information-Channel Decomposition in Electronic Health Record Prediction Models.” Submitted to **KOSMI 2026**. *Under Review*.

- **Han, J.**, et al. “Positivity Diagnostics and Transportability of Prediction Models to Label-Free Target Populations.” Submitted to **KOSMI 2026**. *Under Review*.

TECHNICAL SKILLS

- **Programming:** Python, C++, R, SQL
- **Deep Learning:** PyTorch, scikit-learn
- **Data Analysis:** pandas, NumPy, Matplotlib
- **Tools:** Git, Linux, Postgresql

CERTIFICATIONS AND LANGUAGES

- **Certifications:** SQL Developer (SQLD), ADsP
- **English:** TOEIC 950 / 990